

COMPUTER PROGRAMMING

LAB 7

Lab Title: One-Dimensional Arrays in C++

1. OBJECTIVE

The purpose of this lab was:

- To study the concept of one-dimensional arrays in C++.
- To practice storing multiple values of the same data type in arrays.
- To learn how array elements are accessed through indexing.
- To understand how loops work with arrays for displaying and copying data.
- To explore character arrays and their usage in storing words or messages.

2. THEORY

Introduction to Arrays

An array is a collection of elements of the same data type stored under one variable name. Arrays help programmers manage multiple values efficiently without creating many separate variables. Each element in an array is identified by its position called an index.

In C++, array indexing starts from zero. This means the first element is stored at index 0, the second at index 1, and so on.

Declaring an Array

Arrays are declared by specifying:

- Data type
- Array name
- Number of elements

Example:

```
int values[5];
```

Initializing an Array

Values can be assigned during declaration.

Example:

```
int values[5] = {1, 2, 3, 4, 5};
```

Elements may also be assigned individually later.

Character Arrays

Character arrays are used to store sequences of characters such as words or short messages.

Example:

```
char word[] = "HELLO";
```

Task 1 — Displaying Integer Array Elements

A one-dimensional integer array named `values` was created with five elements. A loop was used to print every element together with its position in the array.

```
#include <iostream>
using namespace std;

int main()
{
    int values[5] = {11, 22, 33, 44, 55};

    for(int index = 0; index < 5; index++)
    {
        cout << "Element at index "
              << index
              << " = "
              << values[index]
              << endl;
    }

    return 0;
}
```

Task 2 — Character Array Traversal

A character array containing the word "ENERGY" was declared. A loop accessed and displayed each character separately.

```
#include <iostream>
using namespace std;

int main()
{
    char word[] = "ENERGY";

    for(int i = 0; i < 6; i++)
    {
        cout << word[i] << endl;
    }
}
```

```
    return 0;
}
```

Task 3 — Transferring Array Data

Two arrays named `firstArray` and `secondArray` were used. Elements from the first array were copied into the second array using a loop.

```
#include <iostream>
using namespace std;

int main()
{
    int firstArray[5] = {2, 4, 6, 8, 10};
    int secondArray[5];

    for(int i = 0; i < 5; i++)
    {
        secondArray[i] = firstArray[i];
    }

    cout << "Copied Elements: ";

    for(int i = 0; i < 5; i++)
    {
        cout << secondArray[i] << " ";
    }

    return 0;
}
```

Task 4 — Finding Total of Array Elements

This program takes five numbers from the user, stores them in an array, and calculates their total sum.

```
#include <iostream>
using namespace std;

int main()
{
    int data[5];
    int total = 0;

    cout << "Enter 5 integers:" << endl;

    for(int i = 0; i < 5; i++)
    {
        cin >> data[i];
    }

    for(int i = 0; i < 5; i++)
```

```
{  
    total = total + data[i];  
}  
  
cout << "Total Sum = " << total << endl;  
  
return 0;  
}
```

5. CONCLUSION

In this lab, the basic concepts of one-dimensional arrays were practiced in C++. Different operations such as declaration, initialization, displaying elements, copying arrays, and working with character arrays were performed successfully. Loops were also used to process array data efficiently. This lab improved understanding of how arrays help in storing and managing multiple values in programming.